

Two pruned proof routes for the $d=4$ fiber conjecture

$$G_\lambda(i) = 0 \iff \lambda = (2, 2)$$

Clio

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Abstract

This is a negative-results note. Two imported-framework routes for the full-support / $d=4$ fiber conjecture are investigated and found *not* to close it: (1) the Foulkes / ω -twisted-plethysm column-antisymmetrization mechanism, which controls only the dominance-extreme (boundary) constituents and has no slot to act on an *ordinary* power ψ^m ; and (2) the eigenvalue-multiplicity framing, which fails on a type mismatch ($G_\lambda(i)$ is a Gaussian integer, multiplicities are non-negative reals) and on a structural mismatch (one conjugacy class vs. a weighted sum over all square-types). Each route is recorded precisely so it is not re-attempted. The portable nugget extracted en route is the ω -derivation of the conjugation symmetry $G_{\lambda'}(i) = i^m \overline{G_\lambda(i)}$. A final synthesis records that the live route is the author's own real-norm machinery.

Setup and the conjecture

Let Λ be the ring of symmetric functions, with h_r the complete homogeneous and e_r the elementary symmetric functions, and let ω be the standard involution, $\omega(h_r) = e_r$. Fix $i = \sqrt{-1}$ and set

$$\psi = h_2 + i e_2 \in \mathbb{C} \otimes \Lambda.$$

For a partition $\lambda \vdash 2m$ define the $d=4$ *fiber value*

$$G_\lambda(i) = \langle s_\lambda, \psi^m \rangle,$$

the Hall-inner-product coefficient of the Schur function s_λ in the degree- $2m$ symmetric function ψ^m . Equivalently, via the character expansion of h_2 and e_2 ,

$$(1+i)^m G_\lambda(i) = \sum_{k=0}^m \binom{m}{k} i^k \chi^\lambda(2^{m-k} 1^{2k}),$$

a weighted sum of irreducible S_{2m} -characters over the involution (“square-type”) classes $2^{m-k} 1^{2k}$.

Conjecture 1 ($d=4$ fiber). For every $\lambda \vdash 2m$,

$$G_\lambda(i) = 0 \iff \lambda = (2, 2).$$

This is verified for all $\lambda \vdash 2m$ with $m \leq 8$ (i.e. $n \leq 16$). The open content is the *full-support* claim: $\text{supp}_{\mathbb{C}}(\psi^m)$ is *every* $\lambda \vdash 2m$ for $m \geq 3$, while for $m = 2$ the only missing shape is the self-conjugate rectangle $(2, 2)$. (Here $\text{supp}_{\mathbb{C}}(\psi^m) = \{\lambda : G_\lambda(i) \neq 0\}$.)

Two routes below were pursued to establish this support statement from an external framework. Both are pruned.

1 The Foulkes / ω -twisted-plethysm route does not reach the interior

We first record the one portable consequence of the ω -involution, then explain why the associated plethysm machinery does not transfer.

Observation 1 (Conjugation symmetry, via ω). Since $\omega(h_2) = e_2$ and $\omega(e_2) = h_2$,

$$\omega(\psi) = \omega(h_2 + i e_2) = e_2 + i h_2 = i (h_2 - i e_2) = i \bar{\psi},$$

where $\bar{\cdot}$ is complex conjugation of coefficients (applied to a real basis of Λ). As ω is a ring homomorphism and conjugation is multiplicative on $\mathbb{C} \otimes \Lambda$,

$$\omega(\psi^m) = (\omega(\psi))^m = (i \bar{\psi})^m = i^m \bar{\psi}^m.$$

Pairing with s_λ and using $\langle s_\lambda, \omega(f) \rangle = \langle \omega(s_\lambda), f \rangle = \langle s_{\lambda'}, f \rangle$ (self-adjointness of ω , $\omega(s_\lambda) = s_{\lambda'}$) together with $\langle s_\lambda, \bar{f} \rangle = \overline{\langle s_\lambda, f \rangle}$ for s_λ real, we obtain

$$G_{\lambda'}(i) = \langle s_{\lambda'}, \psi^m \rangle = \langle s_\lambda, \omega(\psi^m) \rangle = i^m \overline{\langle s_\lambda, \psi^m \rangle} = i^m \overline{G_\lambda(i)}.$$

This conjugation symmetry $G_{\lambda'}(i) = i^m \overline{G_\lambda(i)}$ was already known; the value of Observation 1 is that the ω -twist gives it in two lines, and it is the lens through which the self-conjugate shape $(2, 2)$ becomes the natural candidate vanisher ($\lambda = \lambda'$ forces $G_\lambda(i) = i^m \overline{G_\lambda(i)}$, a phase constraint).

The lead. The route was suggested by MathOverflow #501127 (answers of Wildon and Zaïmi) on the rectangle in a Foulkes-type plethysm: $s_{(k^n)}$ occurs in $h_n[h_k]$ iff k is even. Two proofs are given there.

Remark 1 (Wildon's column-antisymmetrization). Wildon's argument tracks a *single* column-antisymmetrized polytabloid through the surjection $M^{(k^n)} \twoheadrightarrow H^{(k^n)}$ onto the Foulkes module. Antisymmetrizing one column produces a $(-1)^k$ block-swap sign; for k odd the image vanishes, for k even it survives. This decides occurrence of the rectangle $s_{(k^n)}$, which is the *dominance-minimal* constituent of $h_n[h_k]$.

Remark 2 (Zaïmi's ω -twist). Zaïmi uses $\omega(s_\lambda[s_\mu]) = s_\lambda[s_\mu]$ for $|\mu|$ even, reducing the minimal-constituent question, after the twist, to a lex-leading-monomial computation. This is again an argument about a single extremal constituent (now lex-leading after the twist), not the interior.

Proposition 1 (Why the route is pruned). *The Foulkes column-antisymmetrization mechanism can decide the dominance-extreme constituents of $\text{supp}(\psi^m)$ (the boundary of the support polytope) but is not a full-support engine for Conjecture 1. Two obstructions are decisive.*

Discussion. (i) *Extremal, not interior.* Both Remarks 1–2 track *one* extremal constituent through *one* surjection. The $(-1)^k$ cancellation is a statement about the dominance-minimal (resp. lex-leading) polytabloid; it gives no control of an interior λ , for which one would need a per- λ cancellation analysis that is genuinely shape-specific. Indicative of this, the minimal-constituent classification of Paget–Wildon (arXiv:1007.2946) is complete only for $n \leq 5$; there is no uniform interior statement to import.

(ii) *No plethysm slot for the sign to act in.* Both arguments require a genuine *plethysm* / wreath-product induction,

$$h_m[h_2] = \text{ch Ind}_{S_2 \wr S_m}^{S_{2m}} \mathbf{1},$$

so that the $(-1)^k$ block-swap acts in the passage from the permutation module $M^{(2^m)}$ down to the Foulkes quotient $H^{(2^m)}$. But $\psi^m = (h_2 + i e_2)^m$ is an *ordinary product* — a power of a fixed degree-2 element of $\mathbb{C} \otimes \Lambda$ — and lives in the $M^{(2^m)}$ world *before* any quotient to a Foulkes module. There is no plethystic bracket, hence no surjection onto $H^{(2^m)}$ in which the sign mechanism has a slot to act. (The $i e_2$ summand only deforms the ordinary product; it does not manufacture a wreath induction.) Consequently the precise mechanism that proves the rectangle theorem has no analogue here at the interior. $\square$

Conclusion of §1. Foulkes/ ω -plethysm decides the boundary of $\text{supp}(\psi^m)$ only. It does not establish full support, and is pruned as a route to Conjecture 1. The single retained item is the ω -derivation of Observation 1.

2 $G_\lambda(i)$ is not an eigenvalue multiplicity

The lead (GR-1). For an order-4 element $\sigma \in S_{2m}$ and the irreducible representation ρ_λ , the multiplicity of the eigenvalue i in $\rho_\lambda(\sigma)$ is

$$\text{mult}_i(\lambda, \sigma) = \frac{1}{4} \sum_{j=0}^3 i^{-j} \chi^\lambda(\sigma^j).$$

The lead asked whether $G_\lambda(i)$ equals, up to a fixed scalar, $\text{mult}_i(\lambda, \sigma)$ for some natural σ . The relevant references are Staroletov (arXiv:2501.17571), on the spectrum of $\rho_\lambda(\sigma)$, and Amrutha–Prasad–Velmurugan (arXiv:2308.08146), relating an invariant vector to 1 being an eigenvalue.

Proposition 2 (Why the route is pruned). $G_\lambda(i)$ is not (up to a fixed scalar) an eigenvalue multiplicity $\text{mult}_i(\lambda, \sigma)$ of any order-4 element, and neither cited theorem short-cuts Conjecture 1.

Discussion. (i) *Type mismatch.* For every λ and every σ , $\text{mult}_i(\lambda, \sigma) \in \mathbb{Z}_{\geq 0}$ is a nonnegative integer. By contrast $G_\lambda(i)$ is a genuine Gaussian integer that is *generically complex* (nonreal). A nonnegative real cannot equal a generically nonreal quantity up to a fixed scalar, so no identification $G_\lambda(i) = c \cdot \text{mult}_i(\lambda, \sigma)$ can hold with c independent of λ .

(ii) *Structural mismatch.* The multiplicity mult_i is supported on the four powers $\{\sigma^0, \sigma^1, \sigma^2, \sigma^3\}$ of one conjugacy class and uses the 4-cycle character values $\chi^\lambda(4, \dots)$. The fiber value $G_\lambda(i)$, on the other hand, is a $\binom{m}{k} i^k$ -weighted sum over *all* square-types $2^{m-k} 1^{2k}$ with an imaginary sign-twist: it is the projection of s_λ against the *fixed symmetric function* ψ^m , not the class indicator of a single conjugacy class. These are different linear functionals on the character table.

(iii) *Computation.* For all $\lambda \vdash 2m$ with $m \leq 8$ and every natural order-4 element σ , $G_\lambda(i)$ is not proportional to $\text{mult}_i(\lambda, \sigma)$; the zero-sets do not even coincide. The multiplicity mult_i vanishes only at the two one-dimensional representations, whereas $G_\lambda(i)$ vanishes only at $(2, 2)$.

(iv) *The cited theorems do not help.* Staroletov’s result gives the *spectrum* of $\rho_\lambda(\sigma)$ (which roots of unity occur), not the multiplicities; it confirms that i occurs generically but says nothing about the weighted count $G_\lambda(i)$. The Amrutha–Prasad–Velmurugan exceptional pair (relating $\{(2, 2), (3, 1)\}$ to an invariant vector) concerns an order-3 element and the eigenvalue 1, not an order-4 element and i . $\square$

Conclusion of §2. The eigenvalue-multiplicity framing does not capture the invariant $G_\lambda(i)$, and is pruned. Neither Staroletov nor Amrutha–Prasad–Velmurugan short-cuts the $(2, 2)$ conjecture.

3 Synthesis: the proof must come from the norm/Möbius machinery

The imported-framework routes now pruned for the $d=4$ fiber include: CSP carriers; Albion-type z -asymmetric (root-of-unity-twisted) plethysm; the Hermitian family $\Xi = h_2^2 + e_2^2$; the Bethe/spectral home; Foulkes support (§1); and eigenvalue multiplicity (§2). None is a full-support engine.

The *live* route is the author's own hand-built machinery. For two rows,

$$G_{(a,b)}(i) = r_b - r_{b-1}, \quad r_\ell = [u^\ell] (u^2 + (1+i)u + 1)^m,$$

and the real norm $N_\ell = |r_\ell|^2$ is strictly increasing in ℓ with the *single* tie $N_1 = N_2$ occurring only at $m = 2$ (the shape $(2, 2)$). This reduces the two-row instance of Conjecture 1 to one finite inequality — “Lemma 1”, a Möbius-orbit / norm-ratio bound. The standing structural question is whether this real-norm proxy lifts from two rows to general λ : one seeks a Gram-determinant analogue of N_ℓ in which the unique coincidence again pins down the self-conjugate rectangle $(2, 2)$. That is the direction recorded as live.